

THIERRY RUGIRA

Data & AI Engineering | APIs, Pipelines & Automation

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PROFILE

Data and AI engineering practitioner building pipelines, backend APIs and practical automation. Experience spans healthcare utilization systems, environmental data and operational workflows in Rwanda. Focused on trustworthy data, maintainable systems and AI that works within real resource constraints.

EXPERIENCE

SAND Technologies | Kigali, Rwanda

Employee | December 2024 - Present | Work focus: data & AI engineering

- Build and contribute to Python data workflows, Prefect pipelines and PostgreSQL-backed systems, including dbt transformations and data-quality work.
- Develop FastAPI APIs and contribute to AI solutions, support systems and workflow automation, connecting backend services with practical operational needs.
- Contribute to eBuzima healthcare utilization pipelines in a data context of approximately 10,500 rows per day across roughly 450 facilities.
- Work on Rwanda air-quality data in the REMA / PurpleAir context, supporting environmental data workflows and attention to data reliability.

SELECTED SYSTEMS & PRODUCT WORK

eBuzima utilization systems | Professional contribution

Ingestion, transformation and data-quality workflows for healthcare utilization data. Pipeline volume and facility figures describe data scope, not clinical outcomes or adoption.

Waka gym platform | Prototype in development

Systems and product prototyping for gym operations, exploring how operational workflows translate into backend structure and a coherent user experience.

Lightweight medication-safety assistant | Early concept

Exploring a Small AI for Development concept for low-connectivity clinics: trusted medical rules, bounded AI decisions and clinician review. Not deployed or clinically validated.

TECHNICAL SKILLS

Data: Python, SQL, PostgreSQL, Prefect, dbt, data validation

Backend & automation: FastAPI, APIs, backend integration, workflow automation

Systems: AI solutions, system design, product prototyping

EDUCATION

African Leadership University

Bachelor of Software Engineering (BSE), in progress

Current studies: Year 2 | Expected completion: January 2028